Report on Step 3: Advanced Modeling, Interpretation and Conclusions

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02.10.2026

PART I:

Analysis On Complete Dataset

Step 3: Advanced Modeling, Interpretation and Conclusions

EU vehicle CO2 emissions prediction — France | 2024 development data and 2025 temporal test

Pitch: Use bagging/boosting and deep-learning techniques; interpret the models; and derive scientific and business conclusions from the success or failure of the modeling.

Executive Summary

This report presents the final leakage-aware modeling workflow for predicting official WLTP CO_2 emissions (g/km) for French vehicles. The 2024 data were used for development, group-aware validation, feature selection and hyperparameter tuning; the 2025 data were held out as a temporal test set. The report consists of 2 parts, where part 1 represents execution performed on the complete dataset, part 2 represents the execution that was performed on segmented datasets based on fuel modes; namely Mono, PHEV (Plug-in hybrid electric vehicles) and Hybrid.

Executive conclusion part 1: The project progressed from an overly optimistic first benchmark to a leakage-aware final workflow. The strongest conventional model was a tuned ExtraTreesRegressor. On the final workflow, it achieved validation RMSE 6.63 g/km and 2025 test RMSE 10.58 g/km, MAE 5.03 g/km and R^2 0.9625. A neural network was also successful: its best reported test result was RMSE 8.67 g/km and MAE 6.67 g/km. The Lightning MLP had a lower reported RMSE but a higher MAE than the final ExtraTrees model; because the evaluation workflows may not be identical, the results should not be treated as a definitive head-to-head comparison. ExtraTrees remains the primary recommendation because it is the fully validated and interpreted model in the final temporal workflow.

Executive conclusion part 2: The workflow progressed from duplicate consolidation and leakage investigation to a tuned ExtraTreesRegressor. ExtraTrees was selected as the primary conventional model. The extension notebook then trained separate models for Mono-fuel (M), PHEV (P) and Hybrid (H) vehicles, allowing errors and feature importance to be analysed within powertrain regimes.

The segmented ExtraTrees evaluation on 2025 produced MAE 5.0343 g/km, RMSE 10.5761 g/km and R^2 0.9625. Mono and Hybrid models performed substantially better than the PHEV model. The principal weakness was systematic underprediction for unseen petrol/electric PHEV configurations.

1. Data Preparation and Experimental Design

The target is Ewltp (g/km), the official WLTP CO2-emission value. The 2024 French dataset was used for model development, and the 2025 French dataset was held out as a temporal final test. The final notebook reports 15,957 aggregated 2024 records and 14,735 aggregated 2025 records; the final group split contained 12,801 training rows and 3,156 validation rows.

A central improvement was replacing a simple random-style benchmark with a group-aware split. Groups were defined from core technical characteristics: mass, engine capacity, engine power, fuel type and fuel mode. The final split had zero exact core-configuration overlap between training and validation. The 2025 set was not used for model selection or tuning.

Preprocessing was fitted on the training partition only: top-make grouping, group-aware target encoding, frequency maps, IT-code extraction, one-hot encoding and scaling. The final feature design distinguished between a strict design-stage feature set and an extended specification set containing electric range and eco-innovation indicators.

2. Trial-and-error journey

Stage	What was tried	Result and lesson
Initial automated benchmark	LazyPredict comparison of many regressors	Tree ensembles dominated linear models. ExtraTrees, Random Forest, HistGradientBoosting and Bagging were the leading families.
Resource control	Excluded Gaussian-process/kernel-style candidates after estimating a roughly 61.44 GB kernel-matrix requirement versus a 32 GB machine	A practical modeling decision prevented an unproductive or crashing experiment.
First very high scores	ExtraTrees/RF appeared near R^2 0.999 with RMSE around 1.6-1.9 g/km	The result was suspiciously strong, so leakage, target proxies and duplicate configurations were investigated. In the final notebook's broad comparison, Random Forest and ExtraTrees achieved validation RMSE of 1.16 and 1.19 g/km before the stricter feature-set and group-aware tuning stages.
Feature audit	Examined electric range, fuel mode, fuel type, frequency features and possible target-related columns	Electric range was powerful; identifier-derived frequency features were treated cautiously and removed from final feature logic were appropriate.
Feature-set comparison	Strict nine-feature design-stage set versus 17-feature extended specification set	Extended set improved validation RMSE from 9.10 to 7.45 g/km, about 18.1% lower.
Tuning	RandomizedSearchCV with five-fold GroupKFold, only inside the 2024 training data	ExtraTrees achieved the best inner-CV RMSE and the best held-out validation RMSE among tuned candidates.
Deep learning	PyTorch MLP, then an organized PyTorch Lightning version	The network learned the signal but remained less accurate than the tree ensemble on reported test metrics.

3. Ensemble modeling results

The broad comparison showed why nonlinear ensembles were appropriate. The target is influenced by interacting vehicle specifications and categorical powertrain regimes; a linear model cannot represent these interactions as naturally without extensive manual feature construction.

Model/family	Validation RMSE (g/km)	Validation R^2	Interpretation
Random Forest / initial comparison	1.16	0.9995	Best result in the broad automated comparison; later tested under stricter features and group-aware tuning.
ExtraTrees / initial comparison	1.19	0.9995	Very strong early candidate in the broad automated comparison.
LightGBM / initial comparison	1.30	0.9994	Efficient boosting candidate.
Bagging / initial comparison	1.30	0.9994	Strong ensemble baseline in the broad comparison.
Tuned ExtraTrees	6.63	0.9838	Final selected model; best inner-CV RMSE and stronger held-out validation result.
Tuned Random Forest	7.37	0.9799	Close benchmark, but weaker than tuned ExtraTrees.
Extended ExtraTrees feature-set baseline	7.45	0.9795	Outperformed the strict design-stage feature set before tuning.

The selected ExtraTrees configuration was: 678 trees, no maximum-depth limit, max_features approximately 0.8268, minimum samples per leaf 3, minimum samples split 2, random_state 42 and all available CPU cores. The search optimized negative RMSE using five-fold GroupKFold on the training partition. The held-out validation set and the 2025 test set were not used to choose the parameters.

4. Feature engineering and interpretation

The extended specification model used 18 features: vehicle mass, engine capacity, engine power, fuel-type indicators, fuel-mode indicators, electric range and IT-code indicators. The strict model used 10 design-stage variables and excluded electric range and IT features.

Electric range was the most important interpretive finding. In the exploratory audit, the 0-km group had mean emissions around 138.6 g/km, while vehicles with positive electric range averaged around 28.3 g/km. Within the positive-range group, electric range had a strong inverse relationship with emissions: Pearson correlation about −0.786 and Spearman correlation about −0.876. This explains why electric range can produce a major performance improvement, but it is also a potential proxy: it identifies powertrain classes and their emission regimes.

Feature-set comparison showed the trade-off. The strict 10-feature design-stage model achieved validation RMSE 9.10 g/km, while adding electric range and IT indicators reduced validation RMSE to 7.45 g/km, an improvement of approximately 18.1%. Predictive importance is not automatically causal importance.

Interpretability tools used in notebooks included three features of importance, permutation-importance experiments and controlled feature ablation. The main interpretation is global and model-based, not causal claim. For final deployment, SHAP or grouped permutation importance should be added with confidence intervals and evaluated separately for combustion, hybrid and electric subgroups.

5. Deep-learning experiments

The deep-learning notebooks implemented a multilayer perception using 17 selected features. The architecture used 128 → 64 → 32 hidden units, BatchNorm, GELU activations and 5% dropout. Training used AdamW with learning rate 0.001, weight decay 1e−5, MSE loss, ReduceLROnPlateau and early stopping. The direct PyTorch run stopped around epoch 32 after validation loss ceased improving.

Deep-learning version	Reported result	Assessment
PyTorch Lightning MLP	MAE 8.0732; RMSE 11.9008	Lower reported RMSE but higher MAE than final ExtraTrees; workflow comparability should be confirmed before ranking.

The network had approximately 13.2 thousand trainable parameters and used Apple MPS acceleration. The best checkpoint was restored before testing. Reported test metrics were MAE 8.0732 g/km and RMSE 11.9008 g/km. The MLP learned the broad relationship but underperformed the ExtraTrees model.

The result is scientifically useful rather than a failure. Neural networks are not guaranteed to dominate medium-dimensional tabular data, especially when randomized trees capture categorical regimes and threshold-like relationships effectively. The Lightning MLP reported a lower RMSE but a higher MAE than final ExtraTrees; a controlled same-split comparison is required before declaring one model family more accurate overall.

6. Final evaluation and model choice

The final model was set to be the ExtraTreesRegressor model after a 20 iteration randomized search cross validation:

model	Best cv rmse	Validation mae	Validation rmse	Validation r2	best_params
ExtraTrees	7.2341	4.5150	6.6267	0.9838	{'max_depth': None, 'max_features': 0.8268051716 4919, 'min_samples _leaf': 3, 'min_samples _split': 2, 'n_estimators': 678}
RandomForest	7.4048	5.0432	7.3651	0.9799	{'max_depth': 25, 'max_features': 0.4123506965 7748147, 'min_samples _leaf': 2, 'min_samples _split': 9, 'n_estimators': 791}

The 2025 test set contains both familiar and new technical configurations. After retraining on the combined 2024 development data, 83.73% of 2025 rows shared an exact core configuration with the 2024 data, while 16.27% were unseen. Consequently, the overall test score should not be interpreted as pure generalization to entirely new vehicle designs; subgroup reporting for seen versus unseen configurations was investigated.

Subset	Rows	MAE (g/km)	RMSE (g/km)	R^2
All 2025 records	14,735	5.0343	10.5761	0.9625
Seen core configurations	12,337	3.8755	8.6901	0.9733
Unseen core configurations	2,398	10.9959	17.2856	0.9182

Permutation importance was calculated on the held-out 2025 temporal test set. For each feature, its values were randomly permuted ten times while keeping all other features unchanged. Importance was measured as the average increase in RMSE after permutation.

Feature	RMSE increase (g/km)
Electric range (km)	31.87
Fm_P	29.91
m (kg)	18.79
ep (KW)	12.97
ec (cm3)	6.05
Fm_H	2.56
Fm_M	1.82
Ft_petrol	0.90
Ft_diesel	0.71
IT_29	0.71
IT_37	0.70

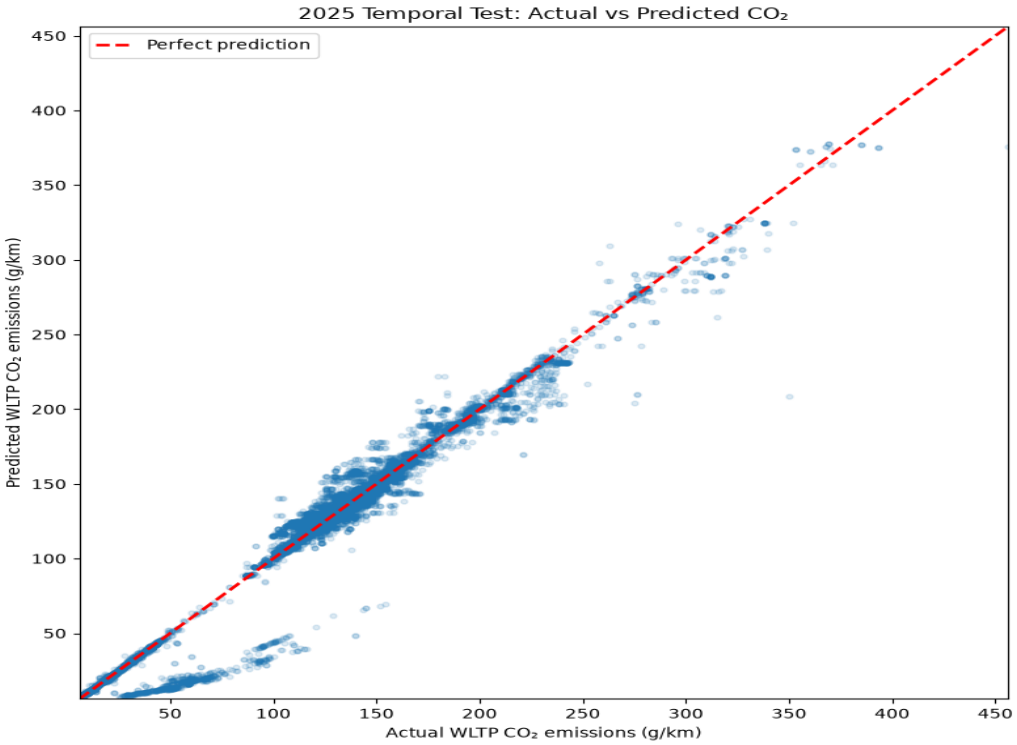
The model relied most strongly on `Electric range (km)`. Randomly permuting this feature increased test RMSE by 31.87 g/km, indicating that electric driving range contains substantial predictive information for WLTP CO_2 emissions.

The next most important variable was `Fm_P`, which indicates the plug-in hybrid powertrain group and increased RMSE by 29.91 g/km when permuted. It is closely related to electric range, so the variables should be interpreted as a correlated PHEV feature group rather than as completely independent effects.

Vehicle mass, engine power, and engine capacity also made substantial contributions. This indicates that the model uses physically meaningful vehicle specifications in addition to powertrain type and electric driving capability.

`IT_38` had no measurable importance, while `Ft_diesel/electric` and `Ft_petrol/electric` had slightly negative importance, suggesting that they provided little incremental predictive value after the other features were included.

Actual vs. predicted CO2 Emissions:



Scatter plot compares the actual and predicted WLTP CO2 emissions for the 2025 temporal test set. The red dashed line represents perfect predictions, where the predicted value is equal to the actual value.

Most observations are located close to the perfect-prediction line, indicating a strong overall agreement between the predicted and observed emissions. The model performs particularly well across the central portion of the target distribution and follows the general upward trend across increasing CO2 values.

However, the plot also reveals several structured error patterns. In the low-emission range, approximately 20–100 g/km, a secondary group of predictions is located substantially below the perfect-prediction line. This indicates systematic underprediction for some low-emission vehicles, potentially due to differences between plug-in hybrid and conventional powertrain configurations.

The model also shows some underprediction for a small number of high-emission vehicles, particularly in the upper tail of the distribution. These errors may be associated with rare high-performance or heavy vehicle configurations that are less represented in the training data.

Overall, the plot confirms that the model captures the main relationship between vehicle specifications and WLTP CO2 emissions. Nevertheless, the remaining structured errors show that overall performance metrics should be complemented with seen-versus-unseen configuration analysis and subgroup-level error analysis.

Residual Analysis:

Residuals were defined as: **residual=actual CO2−predicted CO2**

Therefore, positive residuals indicate underprediction, while negative residuals indicate overprediction.

Residual definition: actual – predicted Residual min: -46.75555434124311 Residual max: 141.1191181349199 Residual mean: 1.1253470707936255 Residual median: -0.13240116693857829	**Largest positive residual:** Actual: 350.0 Predicted: 208.8808818650801 Residual: 141.1191181349199 **Largest negative residual:** Actual: 263.0 Predicted: 309.7555543412431 Residual: -46.75555434124311

The residual distribution showed limited overall bias:

- Mean residual: +1.13 g/km

- Median residual: -0.13 g/km

- Minimum residual: -46.76 g/km

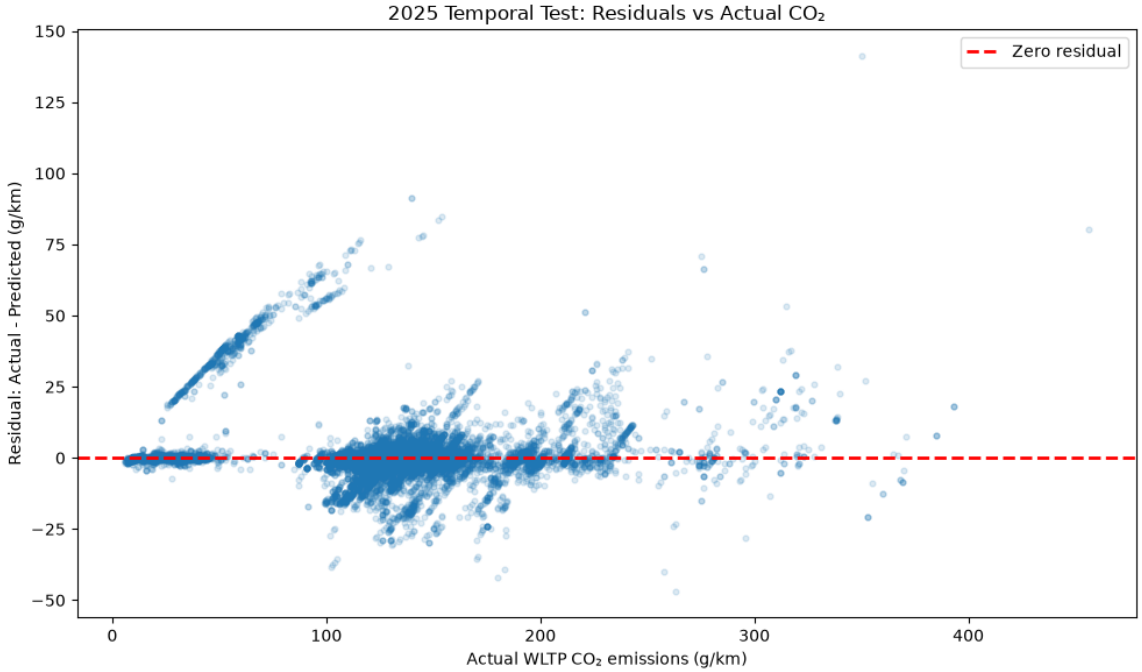
- Maximum residual: +141.12 g/km

The median residual was close to zero, indicating that the model was broadly well calibrated for a typical 2025 record. The slightly positive mean residual was caused by a small number of large underpredictions.

The largest underprediction occurred for a vehicle with an actual WLTP CO2 value of 350.0 g/km. The model predicted 208.88 g/km, resulting in a residual of +141.12 g/km. The largest

overprediction occurred for a vehicle with an actual CO2 value of 263.0 g/km and a prediction of 309.76 g/km, corresponding to a residual of -46.76 g/km.

The residual-versus-actual plot showed that errors were generally concentrated around zero in the central emission range. However, residual spread increased for low-emission and high-emission vehicles. This pattern suggests that the model captures the dominant relationships between vehicle specifications and CO2 emissions, while rare vehicle configurations and unobserved design factors remain difficult to predict.



Error Analysis by Fuel Type and Fuel Mode

Fuel type	Fuel mode	Records	Actual mean (g/km)	Predicted mean (g/km)	Mean residual (g/km)	MAE (g/km)	RMSE (g/km)
petrol/ electric	P	2,466	36.58	27.29	+9.29	9.83	20.93
diesel/ electric	P	229	22.67	13.63	+9.04	9.35	18.05
diesel	H	1,276	147.29	150.12	-2.83	6.38	9.74
petrol	H	3,907	130.63	131.63	-1.00	4.16	6.35
diesel	M	3,121	157.19	157.27	-0.08	3.85	6.24
petrol	M	3,736	147.82	147.99	-0.17	3.06	5.00

Although the model captured that this vehicle had a large engine and very high power, it substantially underpredicted its emissions. This suggests that the model can overgeneralize the low-emission signal associated with PHEV vehicles, especially for rare, high-performance PHEV configurations with short electric range.

A similar pattern appeared for a high-performance PHEV with a 66 km electric range: the actual CO2 value was 140 g/km, while the model predicted 48.47 g/km. This confirms that electric range alone is insufficient to represent the full WLTP CO2 behavior of high-performance PHEV vehicles.

Several heavy petrol/electric PHEV records also had large underpredictions of approximately 77–85 g/km. These records were labelled as seen core configurations because their mass, engine capacity, engine power, fuel type, and fuel mode were present in the 2024 data. However, a seen core configuration does not guarantee an identical full vehicle specification, because additional characteristics such as battery configuration, drivetrain architecture, vehicle version, aerodynamic properties, and WLTP homologation details were not included in the final model.

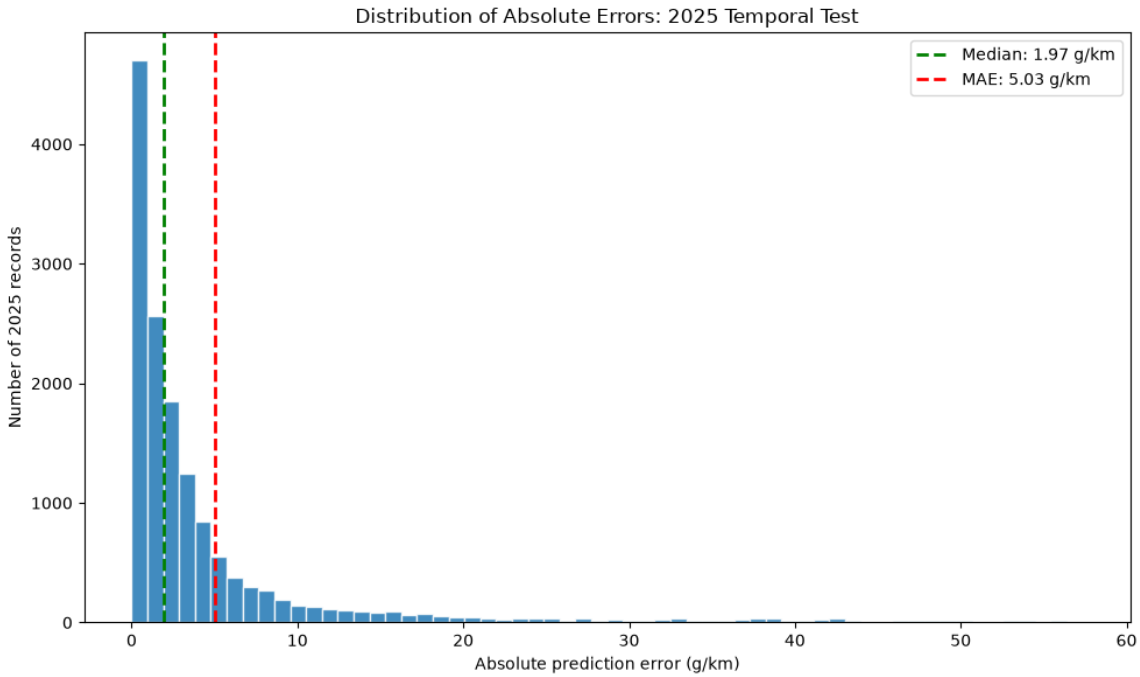
The analysis also identified a large underprediction for an unseen high-emission petrol configuration. Its actual CO2 value was 456 g/km, compared with a prediction of 375.81 g/km. This result indicates that rare, extreme high-emission vehicles are more difficult to predict because comparable observations are less common in the training data.

The repeated records with similar technical specifications should be interpreted as multiple records with related specifications rather than as 20 completely independent vehicle cases. Overall, the largest errors were concentrated in rare high-performance PHEV vehicles and extreme high-emission conventional vehicles, which likely require additional information not available in the leakage-free feature set.

Distribution of Absolute Prediction Errors

Overall regression metrics such as MAE and RMSE summarize model performance with a single value, but they do not show how errors are distributed across individual records.

To better understand the practical accuracy of the final model, the distribution of absolute prediction errors was analyzed on the held-out 2025 temporal test set.



Records with absolute error ≤ 5 g/km: 76.89%

Records with absolute error ≤ 10 g/km: 87.93%

Records with absolute error ≤ 20 g/km: 94.27%

Percentile	Absolute error (g/km)
50th percentile (Median)	1.9700
75th percentile	4.6000
90th percentile	12.2200
95th percentile	22.8700
99th percentile	48.9100
Maximum	141.1200

Error Analysis for Seen/Unseen Vs. Fuel Type/Fuel Mode

The final 2025 test results were analyzed jointly by configuration novelty and powertrain segment.

A record was labelled as **__seen__** when its core technical configuration; defined by vehicle mass, engine capacity, engine power, fuel type, and fuel mode; had already occurred in the combined 2024 training data. Otherwise, it was labelled as **__unseen__**.

Configuration status	Fuel type	Fuel mode	records	Actual mean	Predicted mean	Mean residual	mae	rmse	Median absolute error	Pct error over 20
unseen	petrol/electric	P	404	44.8005	19.7275	25.0731	25.5461	33.5587	31.3815	59.9010
unseen	diesel/electric	P	22	37.9365	17.0100	20.9265	21.1726	30.7355	2.2297	45.4545
seen	petrol/electric	P	2,062	34.9670	28.7708	6.1962	6.7481	17.4150	0.4204	14.2095
seen	diesel/electric	P	207	21.0521	13.2730	7.7791	8.0978	16.1256	0.3335	23.6715
unseen	diesel	H	240	146.5259	150.8798	-4.3540	12.6889	15.5007	12.2451	25.4167
unseen	diesel	M	496	175.8559	175.4246	0.4313	8.2570	11.7162	4.4938	11.2903
unseen	petrol	M	424	164.8132	163.4730	1.3402	5.8056	9.6622	3.3216	6.3679
unseen	petrol	H	812	136.8319	138.6709	-1.8390	7.3638	9.6365	6.5544	2.7094
seen	diesel	H	1,036	147.4675	149.9493	-2.4818	4.9135	7.8278	2.6753	4.5367
seen	petrol	H	3,095	129.0048	129.7805	-0.7757	3.3165	5.1481	1.9260	0.3877

Key Findings: Unseen Petrol/Electric PHEVs Drive Most Large Errors

The joint analysis shows that the high-error tail in the 2025 temporal test set is concentrated overwhelmingly in previously unseen petrol/electric plug-in hybrid (PHEV) configurations.

For unseen petrol/electric vehicles in fuel mode `P`, the model produced a MAE of 25.55 g/km and an RMSE of 33.56 g/km. These vehicles had an average actual CO_2 value of 44.80 g/km, whereas the average prediction was only 19.73 g/km. This corresponds to a mean residual of +25.07 g/km, indicating systematic underprediction.

More than half of these unseen PHEV records (59.90%) had an absolute prediction error above 20 g/km. Their median absolute error was also high at 31.38 g/km, showing that the issue was not caused only by a small number of extreme outliers.

In contrast, seen petrol/electric PHEV configurations had a much lower MAE of 6.75 g/km. However, they still showed a positive mean residual of 6.20 g/km and an RMSE of 17.42 g/km, indicating that PHEVs remain more difficult to predict than conventional petrol and diesel vehicles even when their technical configurations have appeared in the training data.

For conventional petrol and diesel vehicles, prediction performance was substantially more stable. Seen configurations generally achieved MAE values between 2.70 and 4.91 g/km, while unseen conventional configurations achieved MAE values between 5.81 and 12.69 g/km. The largest non-PHEV error occurred for unseen diesel hybrid vehicles, with an MAE of 12.69 g/km and an RMSE of 15.50 g/km.

Therefore, configuration novelty alone does not explain the observed error distribution. The principal challenge is the interaction between configuration novelty and the petrol/electric PHEV segment. New PHEV configurations appear to have CO_2 emission patterns that are not adequately captured by patterns learned from historical data.

7. Scientific conclusions

1. WLTP CO2 emissions are highly predictable from vehicle specifications in this dataset. The strong performance is consistent with the structured physical and regulatory nature of the data, but it does not mean the model has discovered a causal law.

2. Powertrain regime is central. Electric range, fuel type and fuel mode separate low-emission electrified vehicles from predominantly higher-emission combustion vehicles and explain much of the model's predictive power.

3. Group-aware validation materially improves scientific credibility. It prevents repeated or near-identical technical configurations from appearing on both sides of evaluation.

4. The temporal test is essential. The final score is still excellent in 2025, but configuration overlap means a second benchmark with stricter unseen-model or unseen-family splits would provide a harder test.

5. The deep-learning result confirms that model-family choice should follow both data structure and evaluation design. The tested MLP and ExtraTrees showed different error trade-offs, so any final head-to-head decision should use the same split, preprocessing, and reporting metrics.

8. Business conclusions and recommendations

The model can support early CO2 estimation, quality checks and scenario analysis before official WLTP results are available, if input variables are available at prediction time. An MAE around 5.03 g/km on the reported 2025 test is useful for ranking and estimating vehicles, but it should not be treated as a regulatory replacement without subgroup and uncertainty analysis.

Recommended deployment position: use tuned ExtraTrees as the primary prediction model; retain Random Forest as a benchmark alternative; keep the strict design-stage model for early product decisions; and use the extended model only when electric-range and IT specifications are genuinely known before prediction.

Recommended safeguards: report prediction intervals; monitor error by fuel mode, manufacturer, emissions range and unseen configuration status; prohibit post-test variables such as direct target proxies; retrain as new model years arrive; and keep a transparent feature dictionary so business users understand required inputs.

PART II:

Analysis On Segmented Datasets

1. Data Preparation and Experimental Design

The starting cleaned files contained 113,868 2024 records and 88,963 2025 records. Records were grouped by pre-test specification identifiers and technical attributes, and numeric measurements were replaced by registration-weighted means. This reduced the datasets to 15,957 unique 2024 records and 14,735 unique 2025 records.

The target was Ewltp (g/km). The 2024 data were split with GroupShuffleSplit using mass, engine capacity, engine power, fuel type and fuel mode as the core group definition. Training and validation groups were disjoint, so validation measured performance on unseen core technical combinations. The 2025 dataset was kept untouched until final evaluation.

A central improvement was replacing a simple random-style benchmark with a group-aware split. Groups were defined from core technical characteristics: mass, engine capacity, engine power, fuel type and fuel mode. The final split had zero exact core-configuration overlap between training and validation. The 2025 set was not used for model selection or tuning.

Preprocessing was fitted on training data only and then applied to validation and test data. It included make target encoding, training-derived frequency maps, IT-code extraction, one-hot fuel encoding and scaling of continuous variables.

The extension notebook separated the data into **Mono-fuel (Fm=M), PHEV (Fm=P)** and **Hybrid (Fm=H).** Each subset was independently group-split, encoded and scaled. The fuel-mode column was removed because it was constant within each subset.

The 2024/2025 record counts were **Mono** 8,444/6,857, **PHEV** 2,817/2,695 and **Hybrid** 4,696/5,183. Training/validation counts were **Mono** 6,816/1,628, **PHEV** 2,246/571 and **Hybrid** 3,745/951.

Each final subset model used **ExtraTreesRegressor** with n_estimators=388, max_depth=50, max_features=0.6702995511817258, min_samples_leaf=2 and min_samples_split=5, then was retrained on combined 2024 training and validation data.

2. Segmented Temporal Evaluation

The 2025 results show that model performance differs strongly by powertrain segment. Mono and Hybrid predictions were generally stable, whereas PHEV predictions were substantially more difficult, especially for unseen configurations.

Fuel Mode: MONO

2025 subset	Records	MAE	RMSE	R^2
All 2025 records	6857	3.8459	5.9373	0.9746
Seen core configurations	5937	3.4254	5.1633	0.9769
Unseen core configurations	920	6.5589	9.5233	0.9634

Seen configurations: 5,937 / 6,857 (86.58%) Unseen configurations: 920 / 6,857 (13.42%)

Fuel Mode: PHEV

2025 subset	Records	MAE	RMSE	R^2
All 2025 records	2695	9.6732	20.4547	0.5044
Seen core configurations	2269	6.7831	17.1844	0.6289
Unseen core configurations	426	25.0665	32.7719	-0.0684

Seen configurations: 2,269 / 2,695 (84.19%) Unseen configurations: 426 / 2,695 (15.81%)

Fuel Mode: HYBRID

2025 subset	Records	MAE	RMSE	R^2
All 2025 records	5183	4.9229	7.4179	0.9430
Seen core configurations	4131	4.0925	6.3299	0.9541
Unseen core configurations	1052	8.1839	10.6659	0.9125

Seen configurations: 4,131 / 5,183 (79.70%) Unseen configurations: 1,052 / 5,183 (20.30%)

Seen configurations were core technical configurations present in the corresponding 2024 training data. Coverage was 86.58% for Mono, 84.19% for PHEV and 79.70% for Hybrid. The negative R^2 for unseen PHEVs indicates poor generalisation to that subset.

3. Interpretation and Feature Importance

Permutation importance was computed on the 2025 test subsets by repeatedly permuting each feature and measuring the RMSE increase. The dominant features differed by segment.

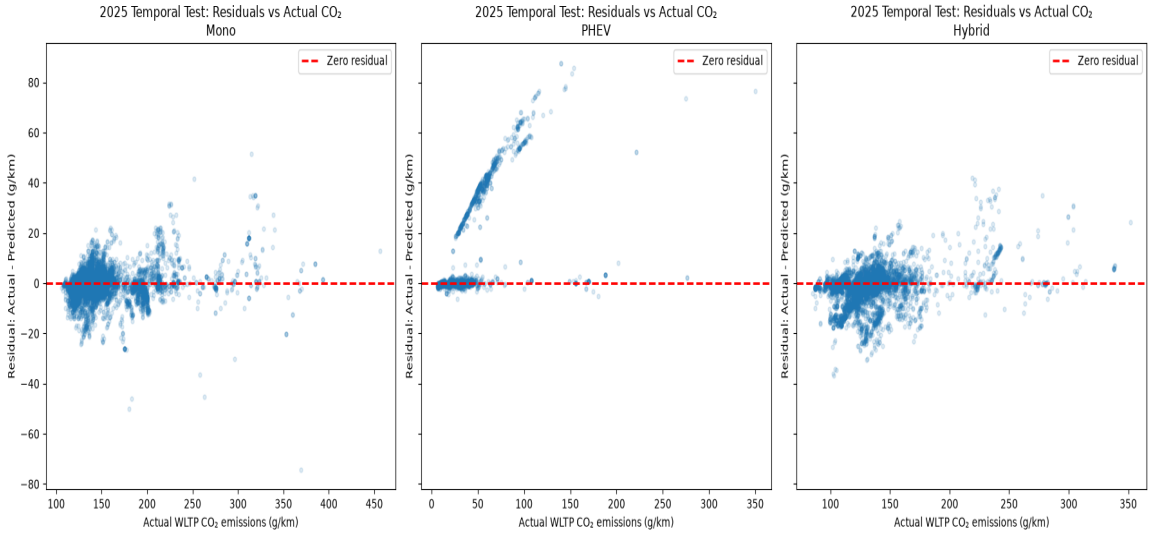
Segment	Most influential features	RMSE increases (g/km)
Mono	Mt, ep (KW), ec (cm3)	26.3757, 14.6895, 11.7005
PHEV	Electric range, ep (KW), Mt	10.4589, 1.8985, 0.3319
Hybrid	ep (KW), Mt, ec (cm3)	14.6276, 10.9463, 7.5671

These are predictive importance measures rather than causal effects. Electric range is strongly associated with PHEV operating and homologation behaviour, but it does not fully describe high-performance PHEVs.

4. Error Analysis

Residual Analysis:

Residuals were defined as actual CO_2 minus predicted CO_2. Positive residuals indicate underprediction and negative residuals indicate overprediction. Mono and Hybrid residuals were mostly concentrated around zero, while PHEV residuals showed a wider spread and many large positive values.

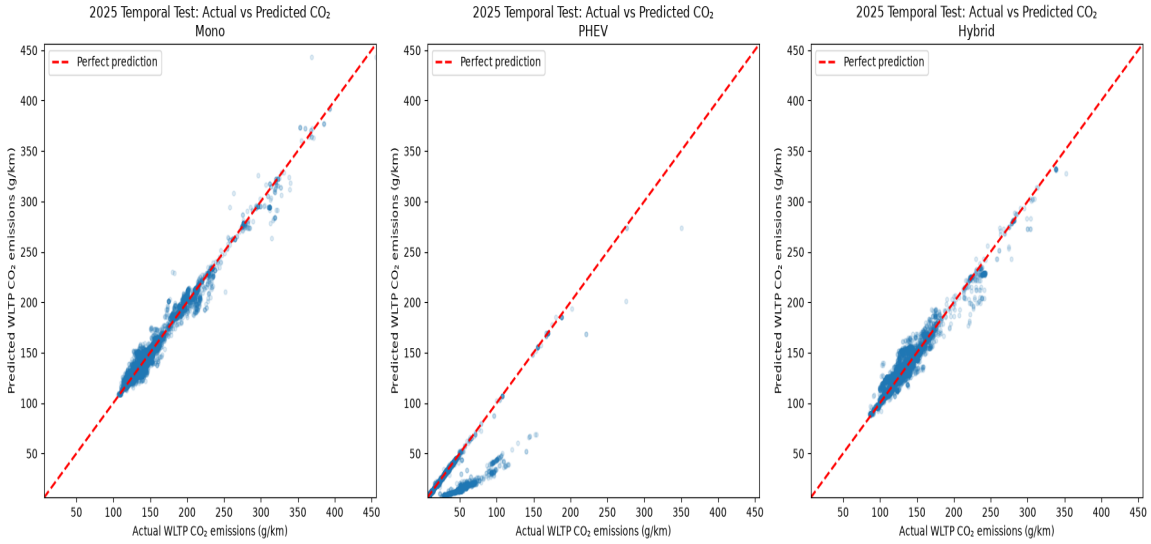


Largest Prediction Errors:

The largest Mono error was 74.16 g/km: actual emissions were 369.0 g/km and the prediction was 443.16 g/km for a seen configuration. The largest PHEV error was 87.44 g/km for an unseen configuration with actual emissions of 140.0 g/km and a prediction of 52.56 g/km. The largest Hybrid error was 41.95 g/km for a seen configuration with actual emissions of 219.0 g/km and a prediction of 177.05 g/km.

PHEV fuel-type analysis confirmed systematic **underprediction**. Petrol/electric PHEVs had mean actual emissions of 36.5780 g/km, mean predictions of 27.3795 g/km, mean residual +9.1985 g/km and RMSE 20.6618 g/km. Diesel/electric PHEVs had mean residual +9.0703 g/km and RMSE 18.0744 g/km.

Actual vs. predicted CO2 Emissions:

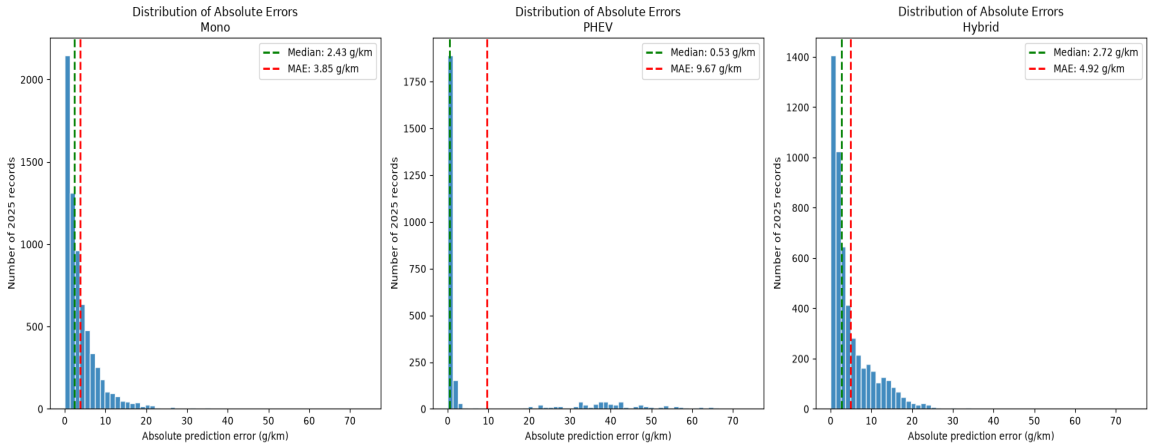


The scatter plot compares the actual and predicted WLTP CO_2 emissions for the 2025 temporal test set. The red dashed line represents perfect predictions, where the predicted value is equal to the actual value.

The Mono and Hybrid models showed strong agreement between actual and predicted WLTP CO_2 emissions, with most observations concentrated close to the perfect-prediction line. Prediction errors increased slightly for high-emission vehicles, where relatively few observations were available.

The PHEV model showed a wider error distribution and a clear tendency to underpredict CO_2 emissions for some vehicles. Several observations with moderate or high actual emissions were predicted substantially below the perfect-prediction line. This result is consistent with the segment-level error analysis, which identified unseen and high-performance petrol/electric PHEV configurations as the main source of large prediction errors.

Distribution of Absolute Prediction Errors:



Model	Metric	Value	Unit
Mono	50th percentile (Median)	2.4300	g/km
Mono	75th percentile	5.1300	g/km
Mono	90th percentile	8.9400	g/km
Mono	95th percentile	12.4400	g/km
Mono	99th percentile	20.8900	g/km
Mono	Maximum	74.1600	g/km
Mono	Error ≤ 5 g/km	74.2700	%
Mono	Error ≤ 10 g/km	92.2100	%
Mono	Error ≤ 20 g/km	98.7200	%
PHEV	50th percentile (Median)	0.5300	g/km
PHEV	75th percentile	2.1200	g/km
PHEV	90th percentile	40.7400	g/km
PHEV	95th percentile	49.4900	g/km
PHEV	99th percentile	65.4000	g/km
PHEV	Maximum	87.4400	g/km
PHEV	Error ≤ 5 g/km	77.0300	%
PHEV	Error ≤ 10 g/km	77.5100	%
PHEV	Error ≤ 20 g/km	78.0000	%
Hybrid	50th percentile (Median)	2.7200	g/km
Hybrid	75th percentile	6.7700	g/km
Hybrid	90th percentile	13.0900	g/km
Hybrid	95th percentile	16.2400	g/km
Hybrid	99th percentile	23.8700	g/km
Hybrid	Maximum	41.9500	g/km
Hybrid	Error ≤ 5 g/km	67.5700	%
Hybrid	Error ≤ 10 g/km	83.8300	%
Hybrid	Error ≤ 20 g/km	97.7600	%

8. Final Conclusions and Recommendations

- WLTP emissions are strongly predictable from vehicle specifications, but performance depends on powertrain regime and configuration novelty.
- Use the extended specification set when electric range and IT indicators are available; retain the strict design-stage set for earlier estimates.

- Mono and Hybrid models generalised well, while unseen PHEV configurations remained the principal weakness.
- PHEV errors indicate that battery capacity, drivetrain architecture, vehicle version, aerodynamics and other homologation-relevant variables may be needed.
- ExtraTrees is the recommended primary model. The MLP is a documented complementary benchmark, not the accuracy leader.
- Always report results by configuration novelty, fuel type, fuel mode and error tail rather than relying only on aggregate metrics.

Recommended safeguards include prediction intervals, monitoring by powertrain and novelty status, periodic retraining with new model years, explicit feature-availability documentation and a stricter future benchmark based on unseen vehicle families or manufacturers.